

Highlights

UL-TransXNet: A Compact ROI Framework for Multi-Dataset Ultrasound Lesion Classification

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- UL-TransXNet classifies ultrasound lesions from expanded ROI crops.
- Three public datasets support multi-dataset in-domain evaluation.
- Validation-only model selection reduces test-set selection risk.
- Automatic ROI experiments connect localization and classification.
- Android experiments quantify prototype accuracy–latency trade-offs.

UL-TransXNet: A Compact ROI Framework for Multi-Dataset Ultrasound Lesion Classification

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Abstract

Ultrasound lesion classification is challenging because speckle noise, weak boundaries, low contrast, and acquisition-dependent variation obscure lesion cues. We present UL-TransXNet, a compact ROI-based TransXNet family architecture for benign-malignant ultrasound lesion classification. Lesion annotations are used to construct expanded ROI crops, after which a parameter/MAC-efficient hierarchical backbone models local texture and broader contextual relationships. We evaluate the model on three public datasets: TN5000 for thyroid nodules, BUSI for breast lesions, and AUL for liver lesions. Under dataset-specific training and the unified ROI reporting protocol, UL-TransXNet obtains mean test AUC/balanced-accuracy pairs of 0.9483/0.8726 on TN5000, 0.8998/0.7961 on BUSI, and 0.8767/0.8200 on AUL. The result does not imply uniform superiority: ConvNeXt-Tiny remains slightly stronger on TN5000 at the selected operating point. Instead, the main finding is multi-dataset in-domain competitiveness, with the strongest average performance across the three datasets and clear gains

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on BUSI and AUL. TN5000 ablations show that the TransXNet family refinements improve operating-point behavior, while MCA contributes mainly to thresholded decision quality and multi-dataset operating behavior rather than to a uniform AUC gain. Automatic-ROI experiments provide workflow evidence: TN5000 verifies detector-front-end feasibility on the primary benchmark, while BUSI/AUL compare oracle ROI, detector-predicted ROI, and full-image inputs and show that automatic crops improve over full-image classification but remain below oracle ROI performance. Android experiments quantify prototype inference trade-offs, where horizontal-flip test-time augmentation improves AUC from 0.9402 to 0.9466 at 226.2 ms per image. These results support UL-TransXNet as a competitive compact ultrasound classifier, while broader external generalization and clinical deployment remain open.

Keywords: Ultrasound image classification, Thyroid nodule, Breast ultrasound, Liver ultrasound, Compact neural network, Mobile inference prototype

1. Introduction

Ultrasound image analysis has become an important pattern recognition problem because ultrasound is widely used in screening, diagnosis support, and follow-up examination for multiple organs. Compared with computed tomography and magnetic resonance imaging, ultrasound is inexpensive, radiation-free, portable, and suitable for repeated acquisition. These advantages make ultrasound attractive for computer-aided diagnosis, but they also make the recognition problem visually difficult. Speckle noise, low con-

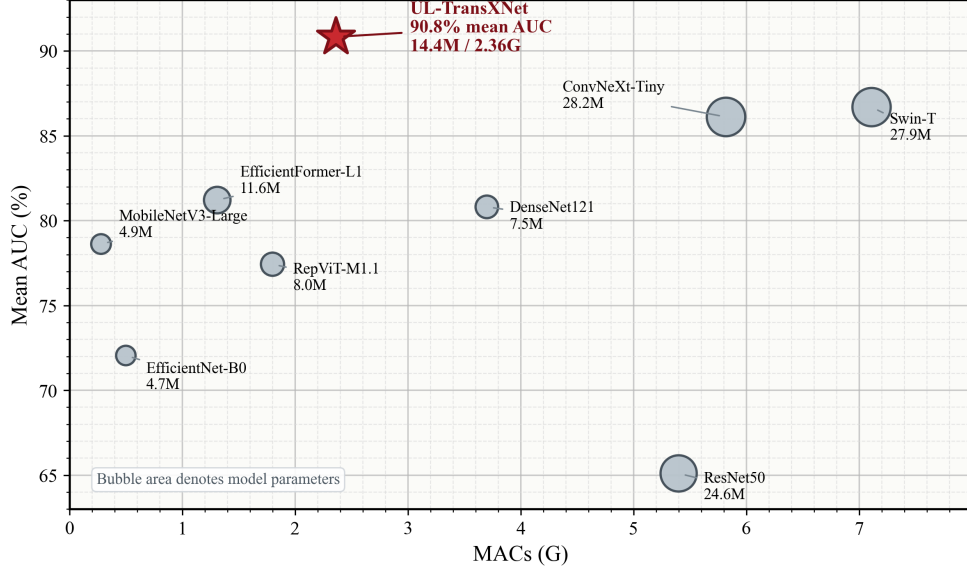


Figure 1: Accuracy–complexity trade-off on the three public ultrasound datasets. The vertical axis reports the mean AUC over TN5000, BUSI, and AUL; the horizontal axis reports MACs for one forward pass; bubble area denotes model parameters.

trast, blurred lesion boundaries, operator dependency, and device-specific appearance shifts can make benign–malignant discrimination substantially harder than standard natural-image classification. A useful ultrasound classifier therefore needs to preserve local lesion details, use surrounding tissue context, and remain compact enough for edge-side feasibility studies.

Public datasets now make it possible to evaluate ultrasound classifiers beyond a single organ. TN5000 provides a large public benchmark for thyroid nodule analysis [1]. BUSI is widely used for breast ultrasound lesion classification [2]. The Annotated Ultrasound Liver (AUL) dataset broadens the setting to liver ultrasound [3]. These datasets are not interchangeable: they differ in anatomy, acquisition appearance, class prevalence, annotation

style, and split protocol. This makes them useful for testing whether an architectural idea is only tuned to one benchmark or remains competitive under different public ultrasound settings.

Recent deep learning methods have improved ultrasound analysis, but three gaps remain important for practical lesion-classification systems. First, many studies report strong results on one dataset while providing limited evidence that the same architecture remains useful under other public ultrasound benchmarks. Second, ROI-based classifiers often rely on annotation-derived crops, which is acceptable for controlled classifier evaluation but leaves the inference-time localization step under-specified. Third, increasingly complex backbones can obscure the practical trade-off between accuracy and computational cost. This trade-off is not a secondary issue in ultrasound: many support scenarios require stable inference under constrained hardware and repeatable preprocessing.

This paper addresses these gaps with UL-TransXNet, a compact ROI framework built from the TransXNet family for ultrasound lesion classification. The method should be understood as a task-specific organization of ROI construction, compact local-global feature modeling, validation-only model-family selection, automatic-ROI workflow evaluation, and mobile prototype measurement. It is not positioned as a new universal medical backbone. TN5000 is treated as the primary large benchmark. BUSI and AUL provide additional dataset-specific in-domain evaluations on breast and liver ultrasound. Structure and module ablations quantify how the TransXNet-family refinements behave under the unified protocol, while automatic-ROI experiments test how much performance is retained when the crop is sup-

plied by a detector rather than by an oracle annotation. A mobile prototype then examines whether the resulting model family can occupy a practical accuracy–latency region.

The main contributions are summarized as follows:

- We present UL-TransXNet as a compact ROI classification framework that combines expanded lesion-centered crops with hierarchical local–global feature modeling for ultrasound benign–malignant discrimination.
- We evaluate the framework on TN5000, BUSI, and AUL under dataset-specific training, and we explicitly separate multi-dataset in-domain competitiveness from stronger but unsupported claims of cross-organ generalization or uniform superiority.
- We audit the final model-family choice with validation-only selection across the three public datasets and provide source-file checksums for the contributing result CSVs, reducing the risk of test-set model selection.
- We report automatic-ROI and Android prototype experiments to connect the controlled ROI classifier to localization and deployment workflow evidence, while keeping detector novelty and clinical deployment claims outside the supported conclusion.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 introduces the proposed method. Section 4 presents the datasets and protocol. Section 5 reports the main experimental findings. Section 6 discusses the implications and limitations, and Section 7 concludes the paper.

2. Related Work

2.1. Ultrasound lesion classification

Ultrasound lesion classification has been studied most extensively in breast and thyroid imaging, where public data availability has enabled repeated benchmarking [2, 1]. Earlier breast ultrasound work often relied on handcrafted morphological and texture descriptors combined with conventional classifiers [4]. More recent deep models have shifted the focus toward representation learning directly from lesion-centered images or regions of interest. For example, Luo *et al.* proposed a multi-view learning strategy for breast ultrasound tumor classification in *Pattern Recognition*, showing that task-specific view construction can improve classification accuracy [5]. These studies confirm that ultrasound lesion classification benefits from architectures that can retain subtle boundary, texture, and contextual cues. However, the majority of published systems remain strongly tied to a single organ or a single benchmark.

Compared with breast ultrasound, public thyroid and liver ultrasound classification benchmarks are less mature but increasingly important. TN5000 has substantially improved the benchmarking environment for thyroid nodule analysis by providing a large-scale public dataset and standardized evaluation protocol [1]. The AUL dataset broadens public benchmarking to liver ultrasound by offering annotated benign, malignant, and normal cases [3]. Taken together, these datasets suggest that a useful ultrasound classification architecture should not only perform well on one organ, but also remain competitive when transferred to datasets with different anatomy, lesion appearance, and acquisition characteristics.

2.2. *Efficient visual architectures*

The design of efficient visual backbones has rapidly evolved from classical convolutional networks to compact CNN–Transformer hybrids. ResNet, DenseNet, EfficientNet, and MobileNetV3 established widely used convolutional design principles such as residual learning, dense feature reuse, compound scaling, and hardware-aware mobile optimization [6, 7, 8, 9]. Vision Transformer and DeiT then demonstrated the strength of attention-based image recognition, while Swin Transformer and ConvNeXt clarified the practical value of hierarchical representations and convolutional inductive biases [10, 11, 12, 13]. These models form the main reference space for modern visual backbones.

Recent lightweight designs focus less on raw parameter reduction and more on where computation should be allocated. MobileViT, EdgeNeXt, EfficientFormer, RepViT, and TransXNet are representative examples of this direction, combining local detail modeling, global context aggregation, and deployment-aware efficiency in different ways [14, 15, 16, 17, 18]. This line of work is particularly relevant to ultrasound analysis, where lesions often occupy a small portion of the image and subtle structural changes can be diagnostically important. Our architecture is built in this efficient-backbone context rather than as a detached medical-only network.

Attention refinements are also relevant to this design space. Differential Transformer introduced differential attention as the difference between two softmax attention maps to reduce attention noise and emphasize relevant context [19]. We do not treat this mechanism as a new primitive. Instead, we adapt it inside a compact ultrasound ROI classifier, where suppressing

common speckle and background responses is useful for lesion-focused discrimination.

Medical image analysis has also begun to adopt lightweight hybrid designs. LightCM-PNet emphasizes real-time ultrasound segmentation under deployment constraints [20], while Lite-MixedNet explores efficient CNN–Transformer fusion for medical image segmentation [21]. These works show that lightweight design is becoming a first-class consideration rather than a secondary implementation detail. Our work follows this trend, but targets ultrasound lesion classification rather than segmentation. The goal is not to propose an entirely separate paradigm, but to derive a coherent and task-motivated architecture that preserves efficiency while improving fine-grained lesion discrimination.

2.3. Multi-dataset validation in medical image analysis

Reporting results on a single benchmark is often insufficient in medical image analysis because performance may depend strongly on dataset-specific factors such as annotation protocol, class prevalence, acquisition device, and institution-specific image characteristics [22, 23]. This issue is especially pronounced for ultrasound, where both image quality and lesion appearance can vary substantially across clinical settings. Recent medical-image-analysis studies have therefore begun to supplement their main benchmarks with robustness analysis, external validation, or deployment-oriented testing [24, 20]. Such evidence does not automatically prove universal generalization, but it gives a more credible picture of where a method is strong and where it remains fragile.

Our paper adopts this perspective. The primary claim is strong dataset-

specific in-domain performance across several public ultrasound datasets, with TN5000 as the main benchmark and BUSI/AUL as additional public benchmarks on breast and liver ultrasound. BUSI and AUL help characterize performance under different anatomy, lesion appearance, and acquisition characteristics, but they are not treated as thyroid-to-breast or thyroid-to-liver external generalization tests. Temperature scaling is used only as validation-derived post-hoc calibration, following the established calibration literature [25]. This positioning keeps the evidence chain aligned with what the experiments can defensibly support.

3. Method

3.1. Problem formulation

Given an ultrasound region-of-interest (ROI) image x , the goal is to predict the lesion label $y \in \{0, 1\}$ for benign–malignant classification, where 0 denotes benign and 1 denotes malignant. The model outputs a two-dimensional logit vector $z = f_\theta(x)$, and the malignant posterior is computed as

$$p(y = 1 \mid x) = \frac{\exp(z_1)}{\exp(z_0) + \exp(z_1)}. \quad (1)$$

The main paper uses the same binary formulation across TN5000, BUSI, and AUL, so that the proposed architecture can be evaluated under a unified lesion-classification setting.

3.2. ROI-based input construction

Figure 2 summarizes the input construction. The lesion annotation defines the initial bounding box. The box is expanded around the lesion center,

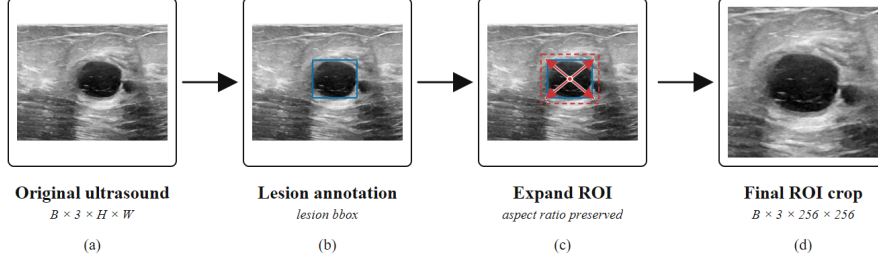


Figure 2: ROI-based input construction. The lesion annotation defines the initial bounding box; an expanded ROI crop centered on the lesion is extracted and resized to the model input size.

clipped to the image boundary, and resized to the model input size. The crop preserves the expanded lesion aspect ratio before resizing; it is therefore an expanded rectangular ROI crop rather than a forced square crop. This procedure keeps the lesion centered while retaining a controlled amount of surrounding tissue context. The main classification protocol therefore uses annotation-derived ROI crops; closed-loop experiments in Section 4 evaluate detector-predicted ROI crops as an auxiliary inference-time front end.

3.3. Overall framework

UL-TransXNet is organized around a restrained workflow assumption: the classifier receives a lesion-centered ROI crop, and the main question is whether a compact local-global backbone can classify that ROI robustly while remaining compatible with automatic ROI generation and mobile inference. The architectural modules below therefore serve this ROI classification workflow. MCA, MUDD, and differential attention are not presented as independent new primitives; they are adaptations and reorganizations of

existing ideas inside a TransXNet-family ultrasound classifier.

Figure 3 illustrates UL-TransXNet, the architecture used in the main experiments. A 256×256 ROI crop is processed by a lightweight four-stage hierarchical backbone. The backbone follows the tiny TransXNet family configuration with stage depths $[4, 4, 15, 4]$, embedding dimensions $[48, 96, 224, 448]$, head numbers $[1, 2, 4, 8]$, and spatial-reduction ratios $[8, 4, 2, 1]$. Each block keeps the central TransXNet idea of local-global token mixing: local convolutional dynamics preserve fine lesion structure, while the global branch models broader contextual relationships.

The backbone is used as a feature extractor. For the custom TransXNet-family models, the final backbone projection is instantiated as a trainable 1000-D feature projection before the task-specific binary head; it is not used as a frozen ImageNet-class logit vector. For timm baselines, the ImageNet classifier is removed and global-pooled features are passed to the same binary head family. The UL-TransXNet task head maps the 1000-D projection to the final binary prediction through a two-layer multilayer perceptron, $1000 \rightarrow 512 \rightarrow 2$, with GELU activation and dropout between the two linear layers. This keeps the classifier head simple and makes the backbone design the main source of representation capacity.

3.4. *Progressive TransXNet family analysis*

Rather than treating the final model as a one-shot design, we organize the architecture as a progressive refinement of the TransXNet baseline [18]. The baseline, denoted **TransXNet**, already combines dynamic local convolution and global attention in a compact hierarchical backbone. The ultrasound setting motivates additional refinements because lesion boundaries can be

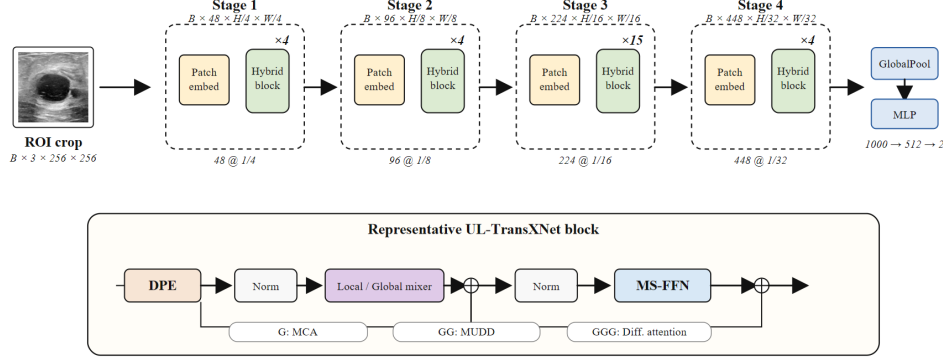


Figure 3: Overview of UL-TransXNet. The ROI crop is resized to 256×256 and processed by hierarchical patch/stem embedding, four TransXNet stages, global pooling, and a lightweight MLP head. The lower panel abstracts the representative block used to describe local/global token mixing, residual fusion, and MS-FFN refinement.

weak, discriminative textures can be local, and useful context can appear at different spatial scales.

The first refinement, **TransXNet-G**, enables multi-scale coordinate attention (MCA), inspired by coordinate attention for lightweight networks [26], in selected global-branch locations. The intent is to improve spatially aware feature selection without a large complexity increase. The second refinement, **TransXNet-GG**, adds multiway dynamic dense connections (MUDD) in deeper stages to encourage short-range feature reuse. The **TransXNet-GGG** variant introduces differential attention in the later global branch. We denote the main reported model as **UL-TransXNet**; in the ablation notation, it corresponds to the **TransXNet-GGG-MCA** configuration. This naming keeps the final model separate from the internal G/GG/GGG identifiers used for ablation. The model-family choice is audited with validation metrics across

the dataset-specific training protocols in Section 5, rather than justified by a single TN5000 test optimum. The TN5000 ablations also show that the intermediate **TransXNet-GG** variant can be the strongest single TN5000 structure. This is an important part of the evidence rather than a contradiction: the refinements improve different operating characteristics, and MCA should be interpreted as an operating-point and multi-dataset behavior module rather than as a guaranteed AUC-improving switch.

3.5. Key modules

Multi-scale coordinate attention (MCA).. When enabled, MCA operates on the global branch after channel splitting. It partitions channels into multiple scale groups, pools each group at a different spatial scale, injects coordinate-aware responses, and generates attention weights for spatially selective feature recalibration. In ultrasound images, lesions may differ through both coarse shape and fine echo patterns; MCA is intended to retain this scale diversity while keeping attention lightweight.

Multiway dynamic dense connection (MUDD).. MUDD is inserted in deeper stages, where semantic features are richer but optimization can be harder. The block feature is divided into multiple channel paths, and a lightweight weight generator computes path-wise aggregation weights from the current activation. Previous block features from the same stage are dynamically fused through these paths. This gives the deeper stages a compact adaptive memory without turning the backbone into a heavy dense network.

Differential attention.. The **GGG** variant adapts differential attention from the Differential Transformer formulation [19], itself building on standard Trans-

former attention [27]. In selected later-stage global-branch locations, queries and keys are split into two groups and two attention maps are computed:

$$A_{\text{diff}} = \text{softmax} \left(Q_1 K_1^\top / \sqrt{d} \right) - \lambda \text{softmax} \left(Q_2 K_2^\top / \sqrt{d} \right), \quad (2)$$

where λ is a learnable balancing coefficient. This module is therefore not claimed as a new attention primitive. In UL-TransXNet, it is used as an ultrasound-specific adaptation inside the global branch to suppress common background or speckle responses and sharpen lesion-related relational contrast when benign and malignant lesions share broad appearance statistics but differ in subtle internal organization.

These modules are introduced as a progressive design family, not as a full factorial decomposition. The ablation study therefore reports both structure-level variants and core-module variants to make the interpretation explicit. The purpose of this analysis is to characterize which operating behaviors change under the current protocol; it does not prove that each module is independently beneficial across every dataset and metric.

3.6. Complexity analysis

Table 1 reports the complexity of the comparison pool and the TransXNet family variants. The numbers are measured with the same binary-classification wrapper used by the training scripts, so they include both the backbone and the lightweight MLP head. All models use 256×256 ROI inputs except EfficientFormer-L1, which follows its 224×224 setting. UL-TransXNet is not the smallest model and is not optimized for local CUDA latency, but it uses fewer parameters and MACs than the strongest TN5000 baselines ConvNeXt-Tiny and Swin-T while preserving stronger multi-dataset

Table 1: Model complexity of the comparison pool and TransXNet family variants. Parameters and MACs are measured for the full binary classifier, including the lightweight MLP head. MACs are reported for one forward pass at the listed input size.

Method	Role	Input	Params (M)	MACs (G)
ResNet50	Prior baseline	256×256	24.56	5.40
DenseNet121	Prior baseline	256×256	7.48	3.70
EfficientNet-B0	Prior baseline	256×256	4.66	0.50
MobileNetV3-Large	Prior baseline	256×256	4.86	0.28
Swin-T	Prior baseline	256×256	27.91	7.11
ConvNeXt-Tiny	Prior baseline	256×256	28.21	5.82
RepViT-M1.1	Prior baseline	256×256	8.04	1.80
EfficientFormer-L1	Prior baseline	224×224	11.62	1.31
TransXNet	Ours / ablation	256×256	13.35	2.32
TransXNet-G	Ours / ablation	256×256	13.42	2.34
TransXNet-GG	Ours / ablation	256×256	14.15	2.38
TransXNet-GGG-noMCA	Ours / ablation	256×256	14.29	2.34
UL-TransXNet	Ours	256×256	14.37	2.36

in-domain behavior on BUSI and AUL.

4. Datasets and Experimental Setup

4.1. Datasets

The main paper uses three public ultrasound datasets: TN5000 for thyroid nodules [1], BUSI for breast lesions [2], and the Annotated Ultrasound Liver (AUL) dataset for liver lesions [3]. TN5000 is the largest benchmark and is used for the most detailed ablation analysis. BUSI and AUL are

Table 2: Overview of the main datasets used in the paper. BUSI and AUL are used in a benign–malignant lesion-classification setting.

Dataset	Organ	Images	B / M	Split protocol	Role
TN5000	Thyroid	5000	1426 / 3574	Official 3500/500/1000 train/val/test	Main
BUSI	Breast	647	437 / 210	Fixed test 129; five training folds on trainval 518	Additional
AUL	Liver	635	200 / 435	Fixed test 127; five training folds on trainval 508	Additional

used to test whether the architecture remains competitive under additional dataset-specific public ultrasound settings. They are not used as external thyroid-to-breast or thyroid-to-liver generalization tests.

4.2. Implementation details

All datasets are processed in an ROI-based manner, but the crop expansion differs across benchmarks. TN5000 and BUSI use a lesion expansion ratio of 0.30, whereas AUL uses a lighter expansion ratio of 0.20. The lesion bounding box is read from XML or manifest metadata, expanded with a minimum crop size of 64 pixels, clipped to the image boundary, and resized to 256×256 ; when a reliable crop is unavailable, the whole image is used as a fallback. The current protocol preserves the expanded box aspect ratio before resizing and does not force a square crop. The only input-size exception in the comparison pool is EfficientFormer-L1, which is evaluated at 224×224 because its wrapper is not compatible with the 256×256 ROI setting used by the other models.

The training pipeline uses the same recipe family across the main runs: random horizontal flip, random rotation ($\pm 4^\circ$), mild affine translation and scaling, brightness/contrast jitter, and random Gaussian blur. These pertur-

bations are kept mild because ultrasound gray-scale intensity, speckle, and gain variation are part of the imaging domain rather than arbitrary color content. Images are loaded as RGB tensors for compatibility with ImageNet-initialized backbones; gray-scale inputs are converted to three channels by the image loader. Evaluation uses deterministic resizing and ImageNet normalization only. All models use AdamW with weight decay 10^{-4} , a lightweight MLP head with dropout 0.3, and label smoothing 0.0. The learning-rate schedule is dataset-specific.

TN5000 uses the official train/val/test split and is reported with three random seeds (17, 27, and 37). The mainline setting uses 70 epochs, patience 12, manual class weights, and differential learning rates of 5×10^{-5} for the backbone and 1.5×10^{-4} for the classification head. BUSI uses a fixed held-out test set together with five training folds on the trainval subset; its main runs use 200 epochs, patience 36, class weighting, and the same $5 \times 10^{-5}/1.5 \times 10^{-4}$ learning-rate pair. AUL likewise uses a fixed held-out test set plus five training folds on trainval, but uses a lighter budget: 20 epochs, patience 4, no class weighting, top-2 checkpoint ensembling, and lower differential learning rates of 1×10^{-5} for the backbone and 3×10^{-5} for the classification head. Thus, BUSI/AUL mean $\pm$ standard deviation values quantify variation across training folds evaluated on a fixed test set, not an outer five-fold cross-validation estimate where each case appears in a test fold exactly once. The mainline scripts enable exponential moving average (EMA) with decay 0.9995; the exception in the baseline pool is BUSI RepViT-M1.1, for which EMA is disabled because the EMA variant produced unstable folds during verification.

For inference-time calibration, each run uses validation-derived temperature scaling [25], averages the top- k validation-improved checkpoints (top-3 on TN5000 and BUSI, top-2 on AUL), applies horizontal-flip test-time augmentation, fits a scalar temperature on validation logits, and searches a decision threshold from 0.10 to 0.95 with step 0.01 using balanced accuracy as the default selection criterion. The resulting validation threshold is applied to the test predictions of that run. Main benchmark tables report mean $\pm$ standard deviation across the three TN5000 seeds or the five BUSI/AUL folds.

Balanced accuracy is used for threshold selection because it gives equal weight to benign and malignant recall and is less tied to class prevalence than raw accuracy. AUC is kept as the threshold-free ranking metric, while macro F1 and accuracy are reported as complementary operating-point summaries.

Importantly, all temperatures and decision thresholds are fixed strictly from validation-derived predictions. No test labels are used for temperature fitting, threshold tuning, or operating-point selection; the test set is used only once for final reporting after these choices are fixed.

4.3. Automatic ROI closed-loop protocol

The main experiments use annotation-derived ROI crops to isolate classifier behavior under a controlled input protocol. To evaluate whether this ROI classifier can be connected to an automatic front end at inference time, we additionally train one-class lesion detectors on BUSI and AUL and perform closed-loop classification with detector-predicted crops. The detector is used only as an auxiliary ROI generator; it is not presented as a novel detection method.

For each BUSI/AUL fold, a one-class YOLO11n detector is trained on the corresponding training annotations and evaluated on the held-out test images. A parallel TN5000 automatic-ROI check trains YOLO11n detectors under three detector seeds and evaluates the official test split. The predicted lesion box with the highest confidence is exported into a VOC-style annotation file and passed through the same expansion, resize, normalization, temperature-scaling, and thresholding code used by the ROI classifier. We then compare three test-time inputs using the same trained UL-TransXNet checkpoints: *oracle ROI*, which uses the ground-truth lesion annotation; *automatic ROI*, which uses the detector-predicted box; and *full image*, which disables the lesion crop and resizes the complete ultrasound image. Thus, the closed-loop experiment tests whether detector-predicted crops can partially substitute annotation-derived ROI construction without retraining the classifier.

4.4. Compared methods

The benchmark pool combines representative CNN, lightweight hybrid, and modern efficient baselines. The comparison set includes ResNet50 [6], EfficientNet-B0 [8], MobileNetV3-Large [9], Swin-T [12], DenseNet121 [7], ConvNeXt-Tiny [13], RepViT-M1.1 [17], and EfficientFormer-L1 [16]. The main paper table reports *Ours* together with the strongest non-*Ours* baseline on each dataset, where “strongest” is determined by mean test AUC under the same reporting convention. The full baseline table is provided in Appendix Table A.21.

4.5. Baseline fairness and reporting convention

All models are evaluated through the same ROI construction, label mapping, train/validation/test splits, metric computation, validation-derived threshold search, temperature-scaling procedure, and test-time augmentation convention unless an exception is explicitly stated. All backbone models use ImageNet-pretrained initialization when available. The classification head is replaced by the binary head used in this study, so ImageNet classifier logits are not used as medical features. The main exceptions are practical and are reported rather than hidden: EfficientFormer-L1 uses 224×224 inputs because its wrapper is not compatible with the 256×256 ROI setting used by the other models; BUSI RepViT-M1.1 disables EMA because the EMA variant produced unstable folds during verification; and AUL uses a lighter training budget than TN5000/BUSI because of the smaller and less stable training setting. These exceptions mean that the benchmark should be interpreted as a carefully matched protocol comparison, not as an exhaustive hyperparameter search for every baseline.

The main results table intentionally reports the proposed model and the strongest non-*Ours* baseline per dataset for readability, while Appendix Table A.21 gives the complete baseline pool. This prevents the main text from implying that a single baseline was chosen in advance as the only comparator. It also makes clear that ConvNeXt-Tiny remains the strongest TN5000 operating-point baseline, whereas UL-TransXNet is selected for its validation-audited three-dataset behavior.

4.6. Evaluation metrics and statistical analysis

The main metrics are AUC, balanced accuracy (BalAcc), macro F1, and accuracy. Unless explicitly stated otherwise, “F1” in the manuscript refers to macro F1. Because the tables emphasize repeated-seed/fold stability, results are reported as mean $\pm$ standard deviation. Thresholded metrics should be interpreted as validation-selected operating points rather than test-tuned oracle points.

As a complementary diagnostic analysis, we also construct case-level summaries from the frozen test prediction CSV files. For TN5000, the three seed predictions are averaged per test case; for BUSI and AUL, the five fixed-test fold predictions are averaged per test case. AUC, sensitivity, specificity, PPV, NPV, ECE, and Brier score are then computed on these averaged predictions, with 95% confidence intervals estimated by bootstrap resampling over test cases. This analysis is used to expose operating-point and calibration behavior on the fixed test sets; it is not treated as prospective clinical significance evidence.

4.7. Local efficiency profiling

Model efficiency is profiled on the TN5000 test split, which provides the largest held-out set among the main datasets. Local CUDA latency is measured on an NVIDIA GeForce RTX 4080 SUPER using batch size 1. Each model is instantiated with the same binary-classification wrapper used for complexity profiling, warmed up for 50 forward passes, and then timed for 300 model-only forward passes with CUDA synchronization. We report the mean model-only latency and the corresponding FPS. A separate TN5000 pipeline pass over 1000 test ROI crops is also logged for diagnosis, but the

main trade-off table uses model-only latency to avoid mixing architecture cost with image decoding and CPU preprocessing overhead.

4.8. Mobile prototype protocol

Mobile inference is evaluated with an Android prototype built on ONNX Runtime 1.24.3. The reported batch experiments were executed on a Xiaomi device (model 24129PN74C, Android 16, arm64-v8a). The prototype study uses TN5000-calibrated UL-TransXNet ONNX exports and compares three inference variants: a single checkpoint, the same checkpoint with horizontal-flip test-time augmentation (TTA), and a three-checkpoint ensemble with TTA. Cold-start runs explicitly cleared the session cache before evaluation, whereas warm-start runs reused the loaded model session. The single-checkpoint mode was measured with one cold-start and three warm-start runs; the TTA and ensemble variants were measured with one cold-start and one warm-start run each. The prototype configuration used lesion expansion ratio 0.30, a fitted temperature of 1.819830, and decision threshold 0.73 inherited from TN5000 validation calibration. We report average end-to-end latency and the corresponding 95th percentile latency over the full ROI preprocessing-plus-inference pipeline.

5. Results

5.1. Main comparison results on public ultrasound datasets

Table 3 summarizes the main comparison results, reported as mean $\pm$ standard deviation across TN5000 seeds or BUSI/AUL training folds evaluated on the fixed held-out test sets. The evidence should be read carefully.

Table 3: Main comparison results on the public ultrasound datasets. For each dataset, the table reports the proposed model and the strongest non-*Ours* baseline ranked by mean test AUC. F1 denotes macro F1.

Dataset	Method	AUC	BalAcc	F1-macro	Acc
TN5000	UL-TransXNet	0.9483 $\pm$ 0.0006	0.8726 $\pm$ 0.0046	0.8503 $\pm$ 0.0123	0.8790 $\pm$ 0.0140
TN5000	ConvNeXt-Tiny	0.9508 $\pm$ 0.0020	0.8794 $\pm$ 0.0023	0.8731 $\pm$ 0.0066	0.8957 $\pm$ 0.0065
BUSI	UL-TransXNet	0.8998 $\pm$ 0.0116	0.7961 $\pm$ 0.0104	0.8018 $\pm$ 0.0119	0.8140 $\pm$ 0.0130
BUSI	Swin-T	0.8190 $\pm$ 0.0091	0.7137 $\pm$ 0.0095	0.6896 $\pm$ 0.0300	0.7194 $\pm$ 0.0445
AUL	UL-TransXNet	0.8767 $\pm$ 0.0009	0.8200 $\pm$ 0.0487	0.7462 $\pm$ 0.0405	0.7685 $\pm$ 0.0418
AUL	Swin-T	0.8319 $\pm$ 0.0117	0.7783 $\pm$ 0.0325	0.7589 $\pm$ 0.0243	0.7717 $\pm$ 0.0211

On TN5000, ConvNeXt-Tiny is slightly stronger than the proposed model at the selected operating point. On BUSI and AUL, however, UL-TransXNet provides a clear gain over the strongest non-*Ours* baseline. Averaged over the three dataset-specific evaluations, the proposed model gives the strongest overall performance, especially in balanced accuracy.

These results delimit the central claim. The proposed model should not be described as uniformly better than every baseline on every dataset. Its advantage is a multi-dataset in-domain pattern: it is competitive on TN5000 and substantially stronger on BUSI and AUL under dataset-specific training. The mean AUC over TN5000, BUSI, and AUL is 0.9083 for UL-TransXNet, compared with 0.8672 for the strongest-per-dataset non-*Ours* rows listed in Table 3. The corresponding mean balanced accuracy is 0.8296 versus 0.7905. This supports the claim that the architecture is a strong compact multi-dataset ultrasound classifier, not a claim of uniform dominance on the primary thyroid benchmark or external cross-organ generalization.

AUL illustrates the distinction between threshold-free ranking and thresh-

Table 4: Claim boundary implied by the current evidence.

Claim	Supported by	Not supported by this study
Multi-dataset in-domain competitiveness	Dataset-specific TN5000/BUSI/AUL tests	External cross-organ transfer
Compact ROI classification	Params/MACs and ROI benchmark results	Smallest or CUDA-fastest model
Validation-audited model selection	Mean validation AUC protocol and checksums	Test-set architecture search
Automatic ROI workflow feasibility	TN5000 detector check; BUSI/AUL auto vs full	Novel lesion detector
Mobile feasibility prototype	Android ONNX Runtime measurement	Clinical deployment validation

olded operating-point metrics. UL-TransXNet has higher AUC and balanced accuracy than Swin-T, whereas Swin-T has slightly higher macro F1 and raw accuracy at its validation-selected threshold. We therefore interpret the AUL result as better ranking and class-balanced operating behavior, not as dominance over every thresholded metric.

Table 6 adds an offline paired statistical check over the same repeated runs used by Table 3. On TN5000, the proposed model is slightly below ConvNeXt-Tiny. On BUSI and AUL, the paired differences are consistently positive for both AUC and balanced accuracy. Because the available repeated units are only three TN5000 seeds and five BUSI/AUL folds, these tests are used as stability evidence rather than as large-sample clinical significance claims.

Table 5 directly compares the original TransXNet baseline with UL-TransXNet under the same dataset-specific reporting protocol. This comparison is important because the method is a TransXNet-family adaptation rather than a detached backbone. The final model is not a TN5000 AUC improvement over the original TransXNet, but it slightly improves the TN5000 validation-selected operating point and gives much larger gains on BUSI and AUL.

Table 5: Original TransXNet versus UL-TransXNet under the same dataset-specific reporting protocol. Δ denotes UL-TransXNet minus TransXNet.

Dataset	TransXNet		UL-TransXNet		Δ	
	AUC	BalAcc	AUC	BalAcc	AUC	BalAcc
TN5000	0.9495	0.8697	0.9483	0.8726	-0.0012	0.0028
BUSI	0.7227	0.6560	0.8998	0.7961	0.1770	0.1401
AUL	0.6582	0.6263	0.8767	0.8200	0.2186	0.1937

Table 7 gives the corresponding complexity–performance view for the full comparison pool. This table makes the trade-off more explicit than the accuracy-only table: UL-TransXNet is parameter/MAC-efficient, but it is not the smallest model and is not CUDA-latency optimal. It achieves the highest mean AUC and mean balanced accuracy with 14.37M parameters and 2.36 GMACs, while its local CUDA batch-1 latency is 30.23 ms, slower than several GPU-optimized baselines. In contrast, the strongest TN5000 baselines ConvNeXt-Tiny and Swin-T require 28.21M/5.82G and 27.91M/7.11G, respectively. The input column is included because EfficientFormer-L1 is evaluated at its native 224 input while the other models use the 256 ROI input used by the proposed model.

5.2. TN5000 progressive-family ablation

Table 8 reports the unified-protocol TN5000 structure ablation. The strongest TN5000 structure is **TransXNet-GG**, which achieves the highest AUC, balanced accuracy, macro F1, and accuracy in this ablation group. The **TransXNet-GGG-MCA** ablation row, which corresponds to UL-TransXNet, is

Table 6: Offline paired statistical comparison between UL-TransXNet and the strongest non-*Ours* baseline in Table 3. Δ denotes UL-TransXNet minus baseline. Confidence intervals are paired bootstrap 95% intervals over seeds or folds; p is an exact two-sided paired sign-flip test. Because only three TN5000 seeds and five BUSI/AUL folds are available, these tests are intended as repeated-run stability checks rather than large-sample clinical significance tests.

Dataset	Baseline	Metric	n	Δ	95% CI	p
TN5000	ConvNeXt-Tiny	AUC	3	-0.0025	[-0.0040, -0.0006]	0.2500
TN5000	ConvNeXt-Tiny	BalAcc	3	-0.0068	[-0.0104, -0.0022]	0.2500
BUSI	Swin-T	AUC	5	+0.0807	[0.0697, 0.0918]	0.0625
BUSI	Swin-T	BalAcc	5	+0.0823	[0.0647, 0.0961]	0.0625
AUL	Swin-T	AUC	5	+0.0449	[0.0375, 0.0574]	0.0625
AUL	Swin-T	BalAcc	5	+0.0417	[0.0160, 0.0674]	0.0625

close but not best on TN5000 alone. This distinction is important because it prevents the paper from over-attributing all gains to the final GGG-MCA setting.

The structure ablation is a progressive-family analysis rather than a full factorial proof. It shows that the TransXNet-family modifications are useful but metric-dependent. The G and GG variants improve thresholded operating-point behavior relative to the baseline. The GGG-MCA configuration is not the TN5000 AUC winner, but it remains competitive and corresponds to the best multi-dataset in-domain pattern in this result set. Table 9 audits the model-family choice using validation AUC from the completed runs. The selection rule is to choose the TransXNet-family row with

Table 7: Complexity–performance trade-off of the comparison pool. CUDA latency is the mean model-only batch-1 latency measured on the local workstation; AUC values are mean test AUCs across TN5000 seeds or BUSI/AUL folds. Mean AUC and Mean BalAcc are averaged over the three datasets.

Method	Type	Input	Params (M)↓	MACs (G)↓	CUDA ms↓	CUDA FPS↑	AUC↑			Mean AUC↑	Mean BalAcc↑
							TN5000	BUSI	AUL		
MobileNetV3-Large	Mobile CNN	256×256	4.86	0.28	3.17	315.1	0.8282	0.7947	0.7357	0.7862	0.7183
EfficientNet-B0	Efficient CNN	256×256	4.66	0.50	6.53	153.1	0.7856	0.7659	0.6099	0.7205	0.6682
EfficientFormer-L1	Hybrid	224×224	11.62	1.31	3.41	293.0	0.9249	0.7917	0.7196	0.8121	0.7330
RepViT-M1.1	Mobile CNN	256×256	8.04	1.80	5.19	192.5	0.8403	0.8028	0.6797	0.7743	0.7020
UL-TransXNet	Ours	256×256	14.37	2.36	30.23	33.1	0.9483	0.8998	0.8767	0.9083	0.8296
DenseNet121	CNN	256×256	7.48	3.70	7.19	139.1	0.9360	0.8154	0.6729	0.8081	0.7401
ResNet50	CNN	256×256	24.56	5.40	3.25	307.5	0.6531	0.7519	0.5481	0.6510	0.6190
ConvNeXt-Tiny	ConvNet	256×256	28.21	5.82	3.30	303.3	0.9508	0.8050	0.8274	0.8611	0.7651
Swin-T	Transformer	256×256	27.91	7.11	6.32	158.3	0.9497	0.8190	0.8319	0.8669	0.7901

Table 8: Unified-protocol structure ablation on TN5000. Mean and standard deviation are computed over three seeds.

Variant	AUC	BalAcc	F1-macro	Acc
TransXNet	0.9495 ± 0.0020	0.8697 ± 0.0100	0.8366 ± 0.0222	0.8640 ± 0.0230
TransXNet-G	0.9481 ± 0.0013	0.8745 ± 0.0074	0.8395 ± 0.0220	0.8660 ± 0.0233
TransXNet-GG	0.9517 ± 0.0011	0.8784 ± 0.0076	0.8552 ± 0.0173	0.8827 ± 0.0170
TransXNet-GGG-noMCA	0.9503 ± 0.0019	0.8660 ± 0.0069	0.8325 ± 0.0013	0.8610 ± 0.0016
TransXNet-GGG-MCA	0.9483 ± 0.0006	0.8726 ± 0.0046	0.8503 ± 0.0123	0.8790 ± 0.0140

the highest mean validation AUC over TN5000, BUSI, and AUL under the dataset-specific training protocols; the source-controlled model-selection protocol records the contributing CSV files and checksums. UL-TransXNet gives the highest mean validation AUC across TN5000, BUSI, and AUL, while GGG-noMCA remains the strongest TN5000 validation-AUC row. Table 10 then shows the corresponding test-set behavior: **TransXNet-GG** is the strongest TN5000-only structure, whereas UL-TransXNet is substantially stronger on BUSI and AUL and therefore gives the better three-dataset average.

Table 9: Validation-performance audit for the TransXNet-family selection. Values are averages over three TN5000 seeds or five BUSI/AUL training folds. The selected row is the final UL-TransXNet configuration used in the main manuscript tables.

Variant	TN5000 val AUC	BUSI val AUC	AUL val AUC	Mean val AUC	Selected
TransXNet	0.9577	0.7796	0.7464	0.8279	no
TransXNet-GG	0.9632	0.8011	0.7390	0.8345	no
GGG-noMCA	0.9700	0.8957	0.8798	0.9152	no
UL-TransXNet	0.9638	0.9199	0.8827	0.9222	yes

5.3. TN5000 core-module ablation

Table 11 isolates the core-module path under the same TN5000 protocol. Differential attention provides the clearest single-module operating-point improvement: **GGG-core+DA** improves balanced accuracy, macro F1, and accuracy over **GGG-core**. MUDD alone gives a smaller gain in thresholded metrics and does not improve AUC in this setting. Adding MCA to the full GGG configuration lowers TN5000 AUC slightly relative to the no-MCA full model, but improves balanced accuracy, macro F1, and accuracy. This supports a restrained interpretation of MCA: it is useful for decision-level robustness and multi-dataset behavior, not a universal AUC booster.

5.4. MCA effect across datasets

Table 12 compares the final GGG model with and without MCA across the three main datasets. The result explains why the final manuscript should keep MCA in the model family but avoid overclaiming it. On TN5000, MCA improves balanced accuracy, macro F1, and accuracy while slightly reducing AUC. On BUSI, MCA provides a large gain across all reported metrics.

Table 10: TN5000-only optimum versus the final multi-dataset configuration. The final reported model is retained for three-dataset behavior, not for the single best TN5000 ablation row.

Dataset	TransXNet-GG		UL-TransXNet	
	AUC	BalAcc	AUC	BalAcc
TN5000	0.9517	0.8784	0.9483	0.8726
BUSI	0.7707	0.7000	0.8998	0.7961
AUL	0.6972	0.6550	0.8767	0.8200
Mean	0.8065	0.7445	0.9083	0.8296

Table 11: Unified-protocol core-module ablation on TN5000. Mean and standard deviation are computed over three seeds.

Variant	AUC	BalAcc	F1-macro	Acc
GGG-core	0.9496 $\pm$ 0.0003	0.8704 $\pm$ 0.0083	0.8324 $\pm$ 0.0136	0.8593 $\pm$ 0.0133
GGG-core+MUDD	0.9470 $\pm$ 0.0006	0.8712 $\pm$ 0.0045	0.8406 $\pm$ 0.0097	0.8687 $\pm$ 0.0111
GGG-core+DA	0.9489 $\pm$ 0.0016	0.8751 $\pm$ 0.0044	0.8490 $\pm$ 0.0100	0.8770 $\pm$ 0.0099
GGG-core+MUDD+DA-noMCA	0.9503 $\pm$ 0.0019	0.8660 $\pm$ 0.0069	0.8325 $\pm$ 0.0013	0.8610 $\pm$ 0.0016
GGG-core+MUDD+DA+MCA	0.9483 $\pm$ 0.0006	0.8726 $\pm$ 0.0046	0.8503 $\pm$ 0.0123	0.8790 $\pm$ 0.0140

On AUL, MCA gives a small AUC and balanced-accuracy gain with nearly unchanged macro F1 and accuracy. Thus, MCA is best described as a module that improves multi-dataset operating behavior, especially on BUSI, rather than as a universally dominant change.

5.5. Automatic ROI localization and closed-loop classification

The main ROI protocol isolates classifier behavior using annotation-derived crops. Figure 4, Table 13, Table 14, Table 15, and Table 16 evaluate

Table 12: Effect of enabling MCA in the final GGG configuration. Mean and standard deviation are computed over TN5000 seeds or BUSI/AUL folds.

Dataset	Variant	AUC	BalAcc	F1-macro	Acc
TN5000	GGG-noMCA	0.9503 $\pm$ 0.0019	0.8660 $\pm$ 0.0069	0.8325 $\pm$ 0.0013	0.8610 $\pm$ 0.0016
TN5000	GGG-MCA	0.9483 $\pm$ 0.0006	0.8726 $\pm$ 0.0046	0.8503 $\pm$ 0.0123	0.8790 $\pm$ 0.0140
BUSI	GGG-noMCA	0.8588 $\pm$ 0.0137	0.7551 $\pm$ 0.0295	0.7526 $\pm$ 0.0300	0.7628 $\pm$ 0.0288
BUSI	GGG-MCA	0.8998 $\pm$ 0.0116	0.7961 $\pm$ 0.0104	0.8018 $\pm$ 0.0119	0.8140 $\pm$ 0.0130
AUL	GGG-noMCA	0.8720 $\pm$ 0.0046	0.8181 $\pm$ 0.0338	0.7462 $\pm$ 0.0283	0.7685 $\pm$ 0.0321
AUL	GGG-MCA	0.8767 $\pm$ 0.0009	0.8200 $\pm$ 0.0487	0.7462 $\pm$ 0.0405	0.7685 $\pm$ 0.0418

a more complete inference chain: an automatic detector first predicts the lesion box, an expanded ROI is constructed from that prediction, and the same trained UL-TransXNet classifiers are then applied without retraining. The TN5000 automatic-ROI checks confirm that the primary benchmark can also be connected to a detector front end and that full-image classification is a much weaker fallback. For BUSI and AUL, the detector does not miss any test image in this evaluation, but localization quality differs by dataset. BUSI reaches a mean IoU of 0.7365, whereas AUL reaches 0.5755 and has a lower high-IoU recall.

Table 14 reports the corresponding TN5000 oracle/automatic/full-image probe from a frozen completed classifier run. The automatic ROI remains close to the oracle ROI and is far stronger than resizing the full ultrasound image, which directly addresses the practical question of whether the annotation-derived ROI protocol can be approximated at inference time. Because this is a single closed-loop probe, it complements rather than replaces the three-seed automatic-ROI summary in Table 13.

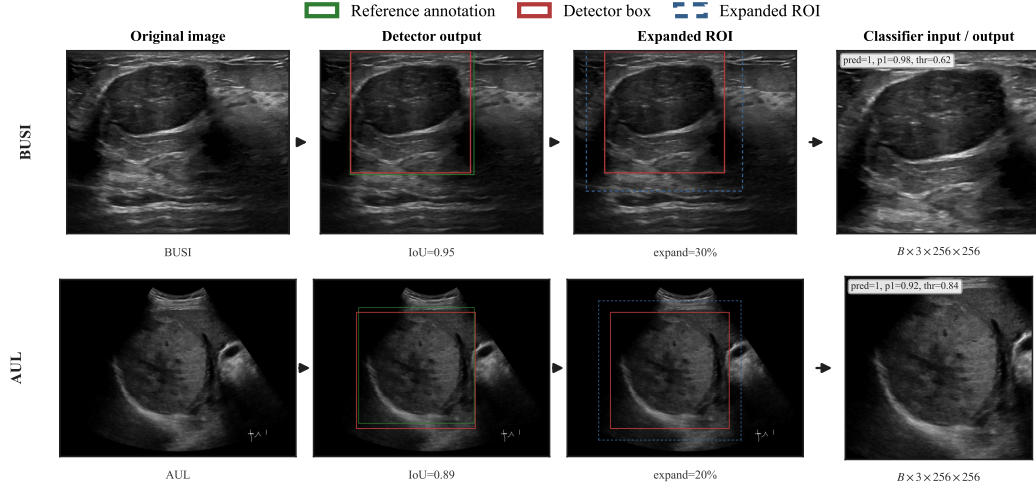


Figure 4: Automatic ROI inference workflow for the BUSI/AUL closed-loop evaluation. Green boxes indicate reference annotations used only for evaluation and visualization; red boxes are detector-predicted lesion boxes; blue dashed boxes are the expanded ROIs used as classifier inputs.

Automatic ROI classification is consistently stronger than full-image classification. On BUSI, the automatic ROI improves AUC by 0.0988 and balanced accuracy by 0.0857 over the full-image input. On AUL, the corresponding gains are 0.1066 AUC and 0.0789 balanced accuracy. The automatic ROI remains below oracle ROI, especially on AUL, which is consistent with the lower localization quality in Table 15. These results support the practical use of an automatic ROI front end, while keeping the detector itself outside the main architectural contribution.

Table 17 adds two completed TN5000 localization-robustness probes. Light box-jitter training and predicted-box mixing preserve the main oracle/automatic/full-image ordering, but neither gives a clear enough automatic-ROI improvement to replace the original classifier. This negative

Table 13: TN5000 automatic ROI sanity check using existing detector and classifier outputs. Detector values summarize three YOLO11n detector seeds; classification values summarize automatic-ROI closed-loop evaluation across the three detector seeds. No classifier retraining is involved.

Block	Metric	Value
Detector	mAP50	0.9520 ± 0.0090
Detector	mean IoU	0.7855 ± 0.0033
Detector	R@0.75	0.7880 ± 0.0122
Detector	no-detection rate	0.0000 ± 0.0000
Auto ROI classifier	AUC	0.9437 ± 0.0042
Auto ROI classifier	BalAcc	0.8686 ± 0.0056
Auto ROI classifier	F1-macro	0.8519 ± 0.0116
Auto ROI classifier	Acc	0.8820 ± 0.0115

result is still useful because it narrows the current claim: the manuscript can support automatic-ROI substitution, but it should not claim that the present training recipe already solves localization-error robustness.

5.6. Qualitative Grad-CAM sanity check

Figure 5 provides a qualitative Grad-CAM sanity check using one trained UL-TransXNet checkpoint per dataset. The maps are generated from late spatial backbone blocks selected for readable ROI-level maps. The activations broadly fall on lesion tissue and adjacent ultrasound context inside the ROI. This visualization is used only as a qualitative sanity check that the classifier is not driven solely by image borders or unrelated background. It is not used as quantitative evidence of causal attention or as a substitute for

Table 14: TN5000 oracle ROI, automatic ROI, and full-image closed-loop probe from frozen existing predictions. This single-probe table complements the three-seed automatic-ROI summary in Table 13; it uses the same trained classifier family and does not involve retraining.

Input	AUC	BalAcc	Sens.	Spec.	F1	Acc
Oracle ROI	0.9464	0.8725	0.8387	0.9062	0.8299	0.8560
Automatic ROI	0.9391	0.8675	0.8522	0.8828	0.8320	0.8600
Full image	0.6490	0.5997	0.7110	0.4883	0.5865	0.6540

Table 15: Detector localization quality used for automatic ROI construction. Values are mean $\pm$ standard deviation across five detector folds.

Dataset	Mean IoU	R@0.50	R@0.75	No det.
BUSI	0.7365 ± 0.0241	0.8403 ± 0.0141	0.6822 ± 0.0548	0.0000
AUL	0.5755 ± 0.0221	0.6740 ± 0.0419	0.4331 ± 0.0201	0.0000

the ablation and closed-loop ROI experiments above.

5.7. Uncertainty and calibration analysis

Table 18 reports bootstrap confidence intervals and calibration metrics for UL-TransXNet using the same seed/fold-level reporting protocol as the main tables. The confidence intervals provide a compact statistical view of the repeated-run stability, while ECE and Brier score quantify probability calibration after the validation-derived temperature-scaling step.

The case-level view in Table 19 and Figure 6 is more clinically interpretable than the seed/fold-only summary. It shows that TN5000 remains well calibrated, while BUSI and AUL have larger reliability error. AUL in

Table 16: Closed-loop classification with oracle ROI, automatic detector ROI, and full-image input. Values are mean $\pm$ standard deviation across five folds. Bracketed intervals are fold-bootstrap 95% confidence intervals for AUC and balanced accuracy.

Dataset	Input	AUC	BalAcc	F1-macro	Acc
BUSI	oracle	0.8998 ± 0.0116 [0.8880, 0.9067]	0.7961 ± 0.0104 [0.7865, 0.8045]	0.8018 ± 0.0119	0.8140 ± 0.0130
BUSI	auto	0.8725 ± 0.0158 [0.8564, 0.8845]	0.7839 ± 0.0179 [0.7704, 0.8010]	0.7893 ± 0.0176	0.8031 ± 0.0167
BUSI	full	0.7737 ± 0.0044 [0.7701, 0.7777]	0.6982 ± 0.0237 [0.6762, 0.7178]	0.6915 ± 0.0378	0.7023 ± 0.0418
AUL	oracle	0.8767 ± 0.0009 [0.8759, 0.8776]	0.8200 ± 0.0487 [0.7755, 0.8599]	0.7462 ± 0.0405	0.7685 ± 0.0418
AUL	auto	0.8507 ± 0.0222 [0.8320, 0.8708]	0.7593 ± 0.0288 [0.7340, 0.7847]	0.6908 ± 0.0561	0.7165 ± 0.0713
AUL	full	0.7441 ± 0.0111 [0.7344, 0.7538]	0.6805 ± 0.0214 [0.6623, 0.6999]	0.6787 ± 0.0203	0.7512 ± 0.0190

particular uses a conservative validation-derived threshold: specificity and PPV are high, but sensitivity and NPV are limited. This supports using the reported probabilities and operating points cautiously, especially under dataset shift.

5.8. Efficiency and mobile prototype

The Android experiment evaluates a TN5000-calibrated prototype of UL-TransXNet. Table 20 and Figure 7 show a trade-off among three inference variants on one device. A single checkpoint provides the lowest end-to-end latency at 116.5 ms per image. Adding horizontal-flip TTA improves both AUC and balanced accuracy, reaching 0.9466 AUC and 0.8762 balanced accuracy at 226.2 ms per image. The three-checkpoint ensemble further increases AUC to 0.9485, but it reduces balanced accuracy at the selected operating threshold and raises latency to 738.6 ms per image. Thus, the single and TTA variants are the practical interactive settings for this prototype, whereas the ensemble is better interpreted as an offline ranking or review option. These measurements are not a full mobile-baseline benchmark.

Table 17: TN5000 localization-robustness probes from completed no-retrain evaluations. The probes test whether light box-jitter training or predicted-box mixing improves the oracle/automatic/full-image operating points. They are kept as diagnostic evidence because neither probe clearly improves the automatic-ROI setting enough to replace the original classifier.

Probe	Input	AUC	BalAcc	F1	Acc
Box-jitter probe	Oracle ROI	0.9510	0.8748	0.8236	0.8480
Box-jitter probe	Automatic ROI	0.9457	0.8739	0.8354	0.8620
Box-jitter probe	Full image	0.6387	0.5923	0.5605	0.6030
Predicted-box mix	Oracle ROI	0.9466	0.8840	0.8511	0.8770
Predicted-box mix	Automatic ROI	0.9429	0.8740	0.8372	0.8640
Predicted-box mix	Full image	0.6598	0.6097	0.5977	0.6670

6. Discussion

The results support a restrained conclusion. UL-TransXNet is a competitive compact ultrasound classifier with favorable multi-dataset in-domain behavior, but it should not be described as uniformly superior on every dataset and every metric. On TN5000, ConvNeXt-Tiny remains slightly stronger in the repeated-seed table; on AUL, Swin-T slightly exceeds UL-TransXNet in macro F1 and raw accuracy despite lower AUC and balanced accuracy. The proposed model becomes clearly advantageous in the multi-dataset view: it is substantially stronger on BUSI, has higher AUC and balanced accuracy on AUL, and has the best average performance across the three public benchmarks under dataset-specific training.

The central story is therefore not that three generic modules are stacked

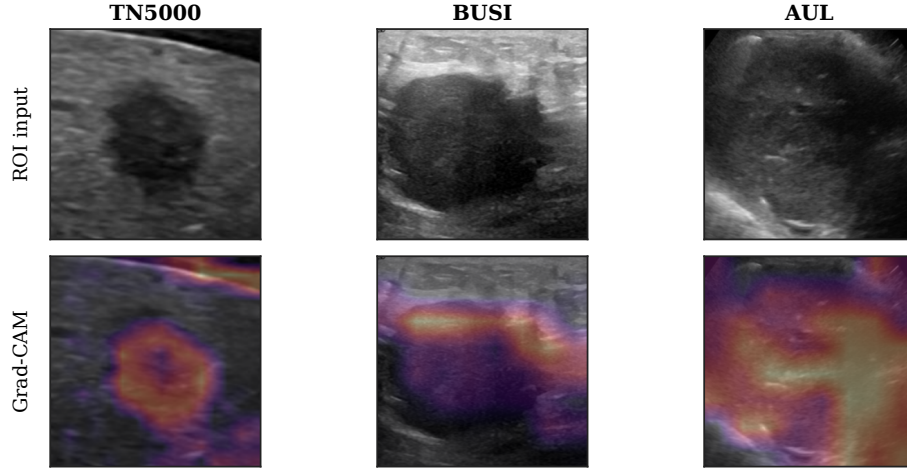


Figure 5: Grad-CAM sanity-check visualization for representative correctly classified ROI inputs. Columns correspond to datasets; the upper row shows the ROI input and the lower row shows the Grad-CAM overlay. The maps are generated from a single UL-TransXNet checkpoint for each dataset and are intended as qualitative localization checks only.

onto TransXNet and happen to improve an average table. A more defensible interpretation is that UL-TransXNet organizes compact local-global feature modeling around an ROI workflow for ultrasound lesions, then tests the resulting classifier under four linked views: controlled oracle ROI classification, validation-audited model-family selection, automatic-ROI substitution, and mobile prototype inference. This framing matches the evidence more closely than a broad claim of a new universal backbone.

The TN5000 ablations clarify the role of the architecture components. The structure-level progression shows that **TransXNet-GG** is the strongest TN5000 variant, while the **TransXNet-GGG-MCA** configuration used by UL-TransXNet is the best three-dataset choice in the validation audit and final result tables. The module-level ablation shows that differential attention

Table 18: Uncertainty and calibration summary for UL-TransXNet on the main test sets. Values follow the manuscript seed/fold-level reporting protocol; intervals are bootstrap 95% confidence intervals over seeds or folds. ECE uses 10 confidence bins.

Dataset	AUC 95% CI	BalAcc 95% CI	ECE	Brier
TN5000	0.9483 [0.9475, 0.9488]	0.8726 [0.8662, 0.8771]	0.0220	0.0741
BUSI	0.8998 [0.8880, 0.9067]	0.7961 [0.7865, 0.8045]	0.0696	0.1358
AUL	0.8767 [0.8759, 0.8776]	0.8200 [0.7755, 0.8599]	0.1283	0.1597

Table 19: Case-level diagnostic statistics for seed/fold-averaged UL-TransXNet predictions on the fixed test sets. Brackets are case-bootstrap 95% confidence intervals. Sensitivity, specificity, PPV, and NPV use the mean validation-derived threshold across the contributing seeds or folds.

Dataset	AUC	Sensitivity	Specificity	PPV	NPV	ECE/Brier
TN5000	0.9540 [0.9381, 0.9676]	0.8750 [0.8511, 0.8981]	0.8828 [0.8406, 0.9209]	0.9559 [0.9397, 0.9704]	0.7085 [0.6571, 0.7576]	0.024/0.068
BUSI	0.9023 [0.8429, 0.9494]	0.7115 [0.5833, 0.8333]	0.8961 [0.8194, 0.9589]	0.8222 [0.7073, 0.9268]	0.8214 [0.7402, 0.8990]	0.104/0.132
AUL	0.8752 [0.8114, 0.9276]	0.7660 [0.6744, 0.8511]	1.0000 [1.0000, 1.0000]	1.0000 [1.0000, 1.0000]	0.6000 [0.4783, 0.7255]	0.104/0.156

is the clearest contributor to TN5000 thresholded metrics. MUDD has a smaller and less uniform effect. MCA does not uniformly improve TN5000 AUC, but it improves TN5000 F1 and accuracy and produces large BUSI gains. This means that MCA should be framed as an operating-point and multi-dataset behavior module rather than as the sole reason for the model family performance.

This interpretation is important in medical image analysis. Ultrasound datasets differ substantially in organ anatomy, acquisition quality, lesion size, and class prevalence. A small AUC difference on one thyroid benchmark is less informative than the pattern across several public benchmarks and

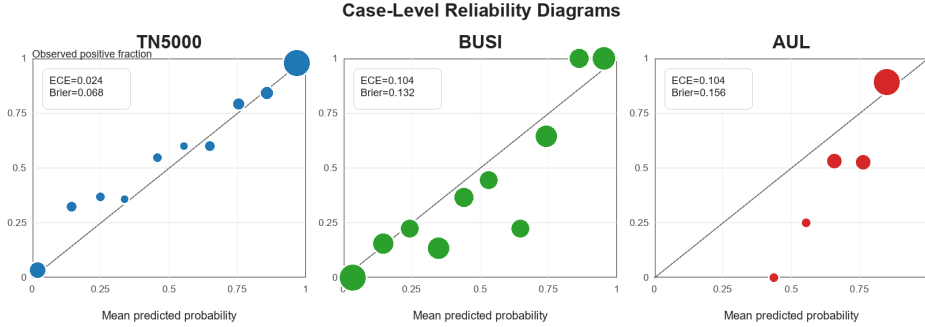


Figure 6: Case-level reliability diagrams for seed/fold-averaged UL-TransXNet predictions. Point size indicates the relative number of test cases in each probability bin. TN5000 is close to the diagonal after temperature scaling, whereas BUSI and AUL show larger calibration deviations.

Table 20: Android prototype results on the TN5000 test set. Latency includes ROI pre-processing and ONNX Runtime inference.

Mode	Runs	AUC	BalAcc	Avg ms	P95 ms
Single	4	0.9402	0.8688	116.5	129.0
+ HFlip TTA	2	0.9466	0.8762	226.2	246.6
3-ckpt ensemble + TTA	2	0.9485	0.8723	738.6	763.5

operating points. The evidence suggests that the proposed architecture is useful because it gives a favorable balance across dataset-specific evaluations, not because every module monotonically improves every metric.

The automatic ROI experiments address a separate practical concern. The main classifier is trained under an annotation-derived ROI protocol, but the TN5000, BUSI, and AUL closed-loop evaluations show that detector-predicted crops can replace oracle crops with a moderate per-

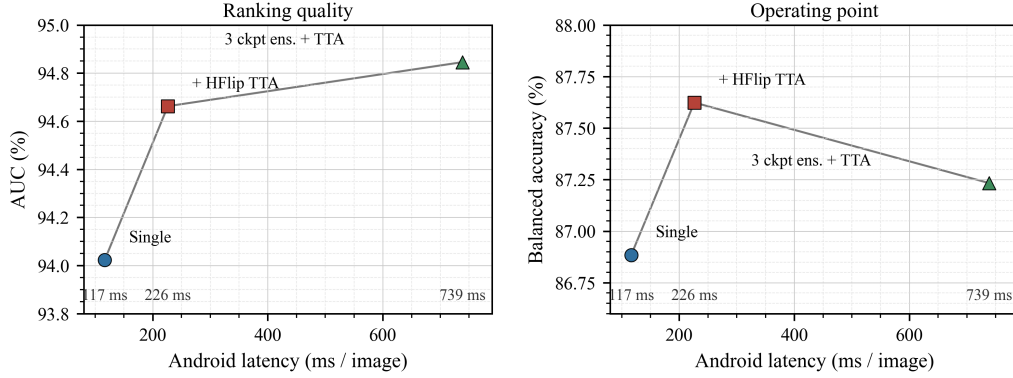


Figure 7: Accuracy-latency trade-off of the Android prototype variants on the TN5000 test set.

formance loss and remain clearly stronger than full-image classification where that comparison is evaluated. The additional TN5000 box-jitter and predicted-box-mix probes are useful negative evidence: they preserve the expected oracle/automatic/full-image ordering, but they do not yet establish a dedicated localization-error-robust training method. This result should be interpreted as workflow evidence rather than detector novelty. In particular, AUL localization is weaker than BUSI localization, which explains part of the remaining gap between automatic and oracle ROI classification.

The Android results remain a practical feasibility check. Many ultrasound classification papers stop at workstation-side accuracy. The mobile experiment shows that a TN5000-calibrated prototype can run in a low-latency mode while preserving strong classification quality, and that a more expensive ensemble mode is available when latency is less critical. These measurements do not constitute clinical deployment validation or prove superiority over all mobile baselines, but they show that UL-TransXNet is compatible

with edge-side inference constraints.

Limitations

This study has several limitations. First, the evaluation is still based on public retrospective datasets, and label definitions, class proportions, and preprocessing conditions are not fully uniform across datasets. BUSI and AUL are dataset-specific in-domain evaluations rather than thyroid-to-breast or thyroid-to-liver external generalization tests. Second, the BUSI/AUL protocol uses five training folds evaluated on a fixed held-out test set, so the reported standard deviations reflect training-fold variation rather than an outer cross-validation estimate. Third, the work focuses on binary benign–malignant classification rather than detection, segmentation, or multi-class clinical diagnosis. Fourth, the automatic ROI experiment uses a practical one-class detector only to form the classifier input; it does not establish a new lesion-detection method, and the AUL results show that localization quality can still limit closed-loop performance. Fifth, case-level reliability analysis shows that BUSI and especially AUL remain less well calibrated than TN5000, so probability outputs should not be interpreted as clinically calibrated risk estimates without further validation. Sixth, the mobile experiment demonstrates prototype feasibility but not a full clinical workflow under realistic operator interaction and acquisition variability. Finally, external robustness remains an open problem because the evaluation is still limited to public retrospective datasets rather than prospective multi-center cohorts. Future studies should therefore focus on broader external validation, stronger detector-classifier integration, calibration under distribution shift, and clinically grounded prospective evaluation.

7. Conclusion

This paper presents UL-TransXNet, a compact ROI framework for benign–malignant ultrasound lesion classification. The evidence supports a restrained but coherent conclusion. UL-TransXNet is competitive on TN5000, clearly stronger on BUSI and AUL, and strongest on average across the three public datasets under dataset-specific training and the unified reporting protocol. The validation audit explains why the final model is selected for three-dataset behavior rather than for the single best TN5000 row. TN5000 ablations show that the architecture gains are not explained by a single module: the GG structure is the strongest thyroid-only variant, differential attention improves thresholded behavior, and MCA mainly improves multi-dataset operating behavior. Automatic ROI experiments further show that detector-predicted crops can connect the classifier to a practical localization front end and remain stronger than full-image classification, including a TN5000 automatic-ROI sanity check for the primary benchmark. Android experiments show that the model can support prototype speed–accuracy trade-offs on mobile hardware. At the same time, broad external generalization and clinical deployment remain unresolved because all reported datasets are public retrospective benchmarks. Future studies should therefore focus on larger external validation, stronger automatic ROI generation, more rigorous calibration under distribution shift, and deployment evaluation under realistic clinical workflows.

Table A.21: Full benchmark across TN5000, BUSI, and AUL using the unified reporting convention. Best per column is shown in bold.

Method	TN5000 AUC	TN5000 BalAcc	BUSI AUC	BUSI BalAcc	AUL AUC	AUL BalAcc
UL-TransXNet	0.9483 $\pm$ 0.0006	0.8726 $\pm$ 0.0046	0.8998 $\pm$ 0.0116	0.7961 $\pm$ 0.0104	0.8767 $\pm$ 0.0009	0.8200 $\pm$ 0.0487
ResNet50	0.6531 $\pm$ 0.0081	0.6319 $\pm$ 0.0067	0.7519 $\pm$ 0.0332	0.6826 $\pm$ 0.0205	0.5481 $\pm$ 0.0771	0.5426 $\pm$ 0.0343
EfficientNet-B0	0.7856 $\pm$ 0.0242	0.7026 $\pm$ 0.0165	0.7659 $\pm$ 0.0368	0.7061 $\pm$ 0.0317	0.6099 $\pm$ 0.1220	0.5958 $\pm$ 0.0912
MobileNetV3-Large	0.8282 $\pm$ 0.0083	0.7626 $\pm$ 0.0082	0.7947 $\pm$ 0.0547	0.7311 $\pm$ 0.0462	0.7357 $\pm$ 0.0062	0.6611 $\pm$ 0.0161
Swin-T	0.9497 $\pm$ 0.0025	0.8782 $\pm$ 0.0051	0.8190 $\pm$ 0.0091	0.7137 $\pm$ 0.0095	0.8319 $\pm$ 0.0117	0.7783 $\pm$ 0.0325
DenseNet121	0.9360 $\pm$ 0.0062	0.8609 $\pm$ 0.0074	0.8154 $\pm$ 0.0278	0.7291 $\pm$ 0.0471	0.6729 $\pm$ 0.0358	0.6304 $\pm$ 0.0706
ConvNeXt-Tiny	0.9508 $\pm$ 0.0020	0.8794 $\pm$ 0.0023	0.8050 $\pm$ 0.0237	0.7261 $\pm$ 0.0420	0.8274 $\pm$ 0.0284	0.6898 $\pm$ 0.0248
RepViT-M1.1	0.8403 $\pm$ 0.0510	0.7603 $\pm$ 0.0427	0.8028 $\pm$ 0.0134	0.7137 $\pm$ 0.0442	0.6797 $\pm$ 0.0029	0.6320 $\pm$ 0.0225
EfficientFormer-L1	0.9249 $\pm$ 0.0021	0.8484 $\pm$ 0.0088	0.7917 $\pm$ 0.0342	0.7213 $\pm$ 0.0226	0.7196 $\pm$ 0.0136	0.6293 $\pm$ 0.0544

Appendix A. Supplementary Benchmark Tables

Table A.21 reports the full benchmark table used to support the main comparison. Values are mean $\pm$ standard deviation across TN5000 seeds or BUSI/AUL training folds evaluated on fixed held-out test sets. The proposed model is UL-TransXNet, which corresponds to the TransXNet-GGG-MCA configuration in the ablation notation; baseline rows are taken from the same comparison protocol.

Appendix B. Reproducibility Note

All main benchmark tables were generated from the seed/fold summary CSV files used by the experiment pipeline. The project archive records the dataset split manifests, ROI construction scripts, training entry points, aggregation scripts, model checkpoints, Android artifacts, validation thresholds, fitted temperatures, prediction CSV files, and a table-level source index required to reproduce the reported TN5000, BUSI, AUL, automatic-ROI, and Android prototype results. The prepared source-controlled release is

intended to include the scripts, compact configurations, split metadata, table summaries, prediction-level CSV files, and the model-selection protocol with source-file checksums that can be shared without redistributing dataset images or large trained weights.

CRedit authorship contribution statement

Zihan Li: Conceptualization, methodology, software, experiments, and writing—original draft.

Cong Yang: Supervision, conceptualization, writing—review and editing, resources, and project administration.

Ethics statement

This study is a secondary analysis of publicly available, de-identified ultrasound datasets. No new patient data were collected by the authors. Ethical approval and consent procedures are governed by the original dataset publications or data providers; therefore, no additional institutional review board approval was required for this retrospective computational study.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work

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Data availability

The datasets analyzed in this study are publicly available from their original sources, which are cited in the manuscript. A reproducibility package is prepared for release at <https://github.com/Afriste/ul-transxnet> and includes source code, split/protocol configuration files, environment specifications, paper-generation scripts, Android demonstration source, artifact-validation utilities, compact result summaries, table provenance, validation thresholds, temperature-scaling values, and prediction CSV files required for table regeneration. Dataset image files, generated ROI image folders, trained checkpoints, detector weights, ONNX binaries, Android application packages, and full raw run logs are not redistributed in the repository. These restricted or large artifacts are retained in the project archive and may be made available from the corresponding author upon reasonable request where permitted by the applicable terms of the original datasets and software providers.

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